

# SARRA BEN DOUB

Data Scientist & AI Engineer · Financial Data · Risk · AI

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## PROFILE

Engineering graduate from the National School of Engineers of Tunis (ENSIT) in applied mathematics and modelling, with experience in Data Science and AI Engineering across financial and insurance environments. I build Machine Learning, Generative AI and intelligent document processing solutions for financial data, risk and analysis problems. I am looking for an opportunity in Data Science, AI Engineering or Machine Learning.

## PROFESSIONAL EXPERIENCE

**BIAT — Financial Document Intelligence & Credit Risk** Feb – Jul 2026

Banque Internationale Arabe de Tunisie · AI Engineer & Data Scientist — Final-year project

- **Document validation & extraction.** Checking the nature of the document, then extracting financial tables from the statements filed as PDFs with the CMF using Docling and OCR (balance sheet, income statement, cash-flow statement), structured and stored in MongoDB.
- **Data cleaning & verification.** Cleaning, normalising and checking the extracted data (OCR errors, inconsistent values), then assigning it to the chart of accounts.
- **Analysis and LLM-generated summary.** Streamlit application (liquidity, solvency, profitability and leverage ratios) then a written summary with levels of assessment, the decision remaining with the analyst. Pipeline orchestrated with n8n, containerised with Docker.

Tech: Python · Docling / OCR · MongoDB · Streamlit · LLM · n8n · Docker

**GAT Assurances — Claims Intelligence & NLP** Jun – Aug 2025

Insurance · AI & Full-Stack Development · Internship

- **Claims-management platform:** public website, customer area and back-office, designed with a graphic designer for the visual identity and user experience.
- **Intent classification** through a French-language rule engine (missing accents, typos), backed by zero-shot classification with BART-large-MNLI when no rule decides.

Tech: Python · Flask · Transformers · BART-large-MNLI · NLP · SQLite · HTML/CSS/JS | github.com/SarraBendoub/insurance-agency-platform

**VALUE — Data Engineering & Analytics** Jul 2024

Data Science · Data Engineering · Internship

- **Multi-source collection** automated with Scrapy and Selenium, cleaning and merging, then **K-Means segmentation** of 314 sites and Folium mapping.

Tech: Python · Scrapy · Selenium · Pandas · scikit-learn · K-Means · Folium | github.com/SarraBendoub/HeritageScope

**BNA Bank — Customer Service Chatbot** Apr – May 2023

Banking · NLP & Machine Learning · Internship

- **Intent classification** on French-language queries: NLTK preprocessing, training and comparison of two architectures, FNN and LSTM, deployed with Flask.

Tech: TensorFlow · Keras · NLTK · Python · Flask | github.com/SarraBendoub/-chatbot\_BNA-main

## PERSONAL PROJECTS — FINANCIAL AI & RISK

**CreditDoc-AI — Extraction from scanned documents & grounded Q&A** 2026

Document AI · OCR · LLM · RAG

- Image preprocessing, OCR and table reconstruction on a scanned financial statement, then extraction to **JSON validated with Pydantic**: label rules first, a local LLM for the missing fields, provenance kept.
- **Grounded RAG** over the document: answers written only from the retrieved passages, page citation, explicit refusal when the information is absent.

Tech: Python · OpenCV · OCR · Ollama · Pydantic · Streamlit

**CreditGuard — Explainable & Monitored Credit Scoring** 2026

Credit risk · Probability of default

- **LightGBM** probability-of-default model, **AUC 0.757**, **SHAP** explainability per prediction and drift monitoring (PSI, Kolmogorov-Smirnov, Chi-squared). Streamlit dashboard, Docker packaging.

Tech: Python · LightGBM · SHAP · SciPy · Streamlit · Docker | github.com/SarraBendoub/CreditGuard

**FraudGuard — Fraud Detection & AML/CFT Monitoring** 2026

Fraud · Financial Crime · Transaction Monitoring

- **XGBoost** fraud detection on **6.4M transactions at 0.13% fraud**: **PR-AUC 0.89**, **F1 0.82**, threshold tuned on the trade-off between missed frauds and false alerts.
- Transaction-monitoring rule engine (velocity, structuring), **BLOCK / REVIEW / ALLOW** action levels, SHAP explainability on alerts, and a **data leakage found and fixed** (PR-AUC 0.9998 obtained from features posterior to the transaction).

Tech: Python · XGBoost · SHAP · pandas · Streamlit | github.com/SarraBendoub/FraudGuard

## TECHNICAL SKILLS

|                                  |                                                                                                                   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Financial Data &amp; Risk</b> | Credit Risk · Probability of Default · Financial Statements · Fraud Detection · Model Monitoring · Explainable AI |
| <b>Machine Learning</b>          | Python · Scikit-learn · XGBoost · LightGBM · SHAP · PyTorch · TensorFlow · Statistics                             |
| <b>Data</b>                      | SQL · PostgreSQL · MongoDB · Pandas · NumPy · Web scraping (Scrapy, Selenium, BeautifulSoup)                      |
| <b>Document AI</b>               | OCR · Docling · Document Processing · LLM                                                                         |
| <b>Generative AI</b>             | RAG · Transformers · LLM · Ollama · Groq                                                                          |
| <b>Deployment</b>                | Docker · FastAPI · Streamlit · n8n · Git                                                                          |
| <b>Reporting &amp; Data Viz</b>  | Power BI · Matplotlib · Seaborn · Streamlit dashboards                                                            |

## EDUCATION

|                                                                                             |             |
|---------------------------------------------------------------------------------------------|-------------|
| Engineering degree — Applied Mathematics & Modelling, National School of Engineers of Tunis | 2023 – 2026 |
| BSc — Data Science, Faculty of Sciences of Tunis                                            | 2020 – 2023 |
| Baccalaureate — Experimental Sciences                                                       | 2020        |

## CERTIFICATION & LANGUAGES

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|---------------------------------------------------------------------------|
| Microsoft Azure AI Fundamentals (AI-900)                                  |
| Arabic native · French fluent · English intermediate (TOEIC) · Italian A1 |